

Donor and Donation Data Analysis Using SQL

SQL Capstone Project

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Tools Used:

- PostgreSQL
- pgAdmin
- SQL

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Executive Summary

This project analyzes donor and donation data using SQL to identify donation patterns, donor behavior, and opportunities for improving fundraising performance. The dataset consists of 1,000 donor records and 1,000 donation records stored across two related tables: Donor_Data and Donation_Data.

The analysis involved data exploration, data quality assessment, data cleaning, and business analysis. Key findings revealed that the organization received a total of 249,085 in donations, with an average donation amount of 249.09. male donors contributed higher total donation amount than female donors despite being fewer in number. Human Resources and Research and Development professionals emerged as valuable donor segments, while California generated the highest total donations among all states.

The findings provide actionable insights that can help improve donor targeting, fundraising strategies, and donor engagement initiatives.

Business Context

Nonprofit organizations and fundraising institutions rely heavily on donor contributions to support their activities and achieve organization goals. Understanding donor behavior is essential for developing effective fundraising strategies, improving donor retention, and maximizing donation revenue.

This project aims to analyze donor demographic information and donation records to uncover trends and patterns in donor behavior. By examining factors such as gender, location, profession, and donation frequency, the organization can identify high-value donor segments and make data-driven decisions to improve fundraising outcomes.

The analysis was conducted using SQL with PostgreSQL and pgAdmin, allowing large datasets to be explored and evaluated efficiently.

Data Overview

The project dataset consists of two related tables:

Donor_Data

This table contains donor profile information, including:

- Donor ID
- Donation Frequency
- University
- Car Ownership
- Second Language
- Favorite Color
- Movie Genre

Donation_Data

This table contains donation and demographic information, including:

- Donor ID
- First Name
- Last Name
- Gender
- State
- Job Field
- Email Address
- Donation Amount

The two tables are linked through the donor ID field, enabling comprehensive analysis of donor characteristics and donation behavior.

Data Dictionary

Column Name	Description
id	Unique identifier for each donor
first_name	Donor's first name
last_name	Donor's last name
gender	Gender of the donor
email	Donor email address
donation	Amount donated by the donor
dontation_frequency	Frequency at which the donor contributes
state	State where the donor resides
job_field	Donor's professional field
university	University attended by the donor
car	Donor's preferred or owned car brand
movie_genre	Donor's preferred movie genre

Methodology

This project was conducted using PostgreSQL and SQL to analyze donor and donation data.

The following steps were followed:

1. Imported the Donor_Data and Donation_Data datasets into PostgreSQL using pgAdmin.
2. Explored the datasets to understand the structure, number of records, and available variables.
3. Assessed data quality by checking for missing values, duplicate records, and data inconsistencies.
4. Performed data cleaning and validation checks to ensure the accuracy of the analysis.

5. Conducted exploratory data analysis using SQL queries, aggregation functions, filtering, grouping, and sorting techniques.
6. Answered eight business questions to identify patterns and trends in donor behavior.
7. Generated insights and recommendations based of the findings.

Data Exploration Findings

Dataset Overview

The project dataset consists of:

- **Donor Data:** 1,000 donor profile records.
- **Donation Data:** 1,000 donation records.

The two tables are linked through the id column.

Summary Statistics

METRIC	VALUE
Total Donors	1,000
Total Donation Records	1,000
Total Donation Amount	249,085.00
Average Donation	249.09
Minimum Donation	5
Maximum Donation	500

Gender Distribution

Gender	Count
Female	508
Male	492

Observation

The donor population is nearly evenly split between male and female donors, with females representing a slight majority (50.8%).

Top 10 Donors

Rank	Donor Name	Donation
1	Beverlie Andriesse	500
2	Wallie Leather	500
3	Peder Rilton	499
4	Clevie Camilletti	499
5	Worthy Le Feaver	498
6	Amalea Knill	497
7	Corbett Lansdale	494
8	Nathaniel McGenn	494
9	Tonnie Stockney	494
10	Corbin Rawne	493

Key Insights

1. The dataset contains 1,000 donor records and 1,000 donation records, indicating a complete one-to-one relationship between donors and donations.
2. The organization received a total of 249,085 in donations.
3. The average donation amount is 249.09.
4. Donations range from 5 to 500, showing a widespread in donor contributions.
5. The highest donation recorded is 500.

6. Female donors slightly outnumber male donors.
7. Several donors contributed amounts very close to the maximum donation value, suggesting a strong group of high-value donors.

Data Quality Assessment

A data quality assessment was conducted to evaluate the completeness, consistency, and reliability of the dataset before performing further analysis.

Missing Values Analysis

The Donation Data table was assessed for missing values in key fields including first name, last name, email address, and donation amount.

Column	Missing values
first_name	0
last_name	0
email	0
donation	0

The results indicate that the Donation Date table contains no missing values in the key fields analyzed.

The Donor Data table was also assessed for missing values.

Column	Missing Values
Donation frequency	0
University	459
Car	0
Movie genre	0

The analysis revealed a significant number of missing values in the university column, with 459 records lacking university information. All other assessed columns were complete.

Duplicate Record Assessment

Duplicate checks were performed using the donor ID field in both tables.

No duplicate records were identified in either Donor Data or Donation Data. This confirms that each donor is represented only once in the dataset.

Donation Range Validation

Donation values were reviewed to identify any unusual or invalid entries.

Metric	Value
Minimum Donation	5
Maximum Donation	500

The donation values fall within a reasonable range and no invalid donation amount were detected.

Data Quality Conclusion

Overall, the dataset demonstrates a high level of quality. The only notable issue identified was the presence of missing values in the university field. Aside from this, the dataset contains no missing donor information, no duplicate records, and valid donation values, making it suitable for further analysis.

Data Cleaning

Following the data quality assessment, a review of missing values identified 459 missing records in the university column within the Donor Data table.

Missing University Values

Metric	Value
Missing University Records	459

Since the university field is not a critical identifier and more than 45% of the records are missing, the records were retained to avoid unnecessary data loss.

For analytical purposes, missing university values will be treated as “Unknown” when required in reporting and segmentation analyses.

Data Cleaning Summary

The following data quality issues were addressed:

- Verified that there were no missing values in key donor and donation fields.
- Confirmed the absence of duplicate records.
- Validated that donation amounts were within an acceptable range
- Identified and documented missing university values
- Retained all records to preserve the integrity and completeness of the dataset.

The dataset was deemed suitable for business analysis and reporting.

Business Analysis and Findings

Business Question 1: which Gender Contributes the Most Donations?

To understand donor contribution patterns, donations were analyzed by gender.

Gender	Donor Count	Average Donation	Total Donation
Male	492	259.41	127,628
Female	508	239.09	121,457

Findings

- Female donors represent the majority of donors, accounting for 508 individuals.
- Male donors account for 492 individuals.
- Despite having fewer donors, male donors contributed a higher total donation amount.
- The average donation from male donors (259.41) exceeded the average donation from female donors (239.09).

Business Insight

Male donors generate a larger share of total donations and contribute higher average donation amounts. This suggests that gender-based donor segmentation may be useful when designing future fundraising campaigns.

Business Question 2: Which Job Fields Contribute the most Donations?

Donation were analyzed by job field to identify the most valuable donor segments.

Job Field	Donor Count	Average Donation	Total Donation
Human Resources	93	247.96	23,060
Research and Development	84	272.17	22,862
Product Management	90	253.31	22,798
Business Development	94	236.87	22,266
Engineering	93	236.22	21,968
Training	84	258.58	21,721
Accounting	80	256.30	20,504
Services	80	248.23	19,858
Support	79	246.52	19,475
Sales	83	229.02	19,009

Findings

- Human Resources generated the highest total donation amount (23,060).
- Research and Development donors recorded the highest average donation (272.17)
- Product Management and Business Development also contributed significantly to total donations.
- Sales donors generated the lowest total donations among the top ten job fields.

Business Insight

Job field appears to influence donation behavior. Donors working in Human Resources, Research and Development, and Product Management represent particularly valuable donor segment and may be targeted for future fundraising campaigns and donor engagement initiatives.

Business Question 3: which States Generate the Highest Donation?

Donations were analyzed by state to identify the geographic regions contributing the highest donation amounts.

State	Donor Count	Total Donation	Average
California	113	30,264	267.82
Texas	95	24,097	253.65
Florida	90	20,562	228.47
New York	58	14,759	254.47
Virginia	39	10,750	275.64
Illinois	34	8,674	255.12
District of Columbia	30	8,376	279.20
Tennessee	30	8,316	277.20
Georgia	33	8,046	243.82
Ohio	32	6,876	214.88

Findings

- California generated the highest total donation amount (**30,264**).
- Texas ranked second with total donations of **24,097**.
- Florida ranked third with **20,562** in donations.
- The **District of Columbia** recorded the highest average donation (**279.20**) among the top 10 states.
- Ohio had the lowest average donation among the top 10 states (**214.88**).

Business insight

Donation performance varies significantly by location. California, Texas, and Florida represent the strongest fundraising markets based on total donations. The District of Columbia and Tennessee show particularly high average donation values, suggesting that donors in these locations may be strong candidates for premium donor engagement programs.

Business Question 4: How Frequently Do Donors Give?

To understand donor behavior, donors were grouped according to their donation frequency.

Donation Frequency	Donor Count
Yearly	135
Never	133
Weekly	129
Once	128
Daily	128
Monthly	118
Seldom	118
Often	111

Findings

- Yearly donors represent the largest donor group with **135 donors**.
- Weekly, daily, and one-time donors each account for more than 120 donors.
- A significant number of donors (**133**) are classified as “Never”, indicating potential inactive donors.
- “Often” donors represent the smallest group with **111 donors**.

Business Insight

The donor base consists of a mix of recurring and infrequent donors. The large number of yearly donors suggests that annual fundraising campaigns may be particularly effective. Additionally, the presence of inactive donors (“Never”) presents an opportunity for re-engagement campaigns aimed at increasing participation and donation frequency.

Advanced Analysis

Business Question 5: how does Donation Frequency Differ by Gender?

A join was performed between the Donor Data and Donation Data tables to analyze donation frequency patterns by gender.

Gender	Donation Frequency	Donor Count
Female	Never	73
Female	Weekly	73
Female	Once	67
Female	Monthly	64
Female	Yearly	63
Female	Daily	61
Female	Seldom	56
Female	Often	51
Male	Yearly	72
Male	Daily	67
Male	Seldom	62
Male	Once	61
Male	Often	60
Male	Never	60
Male	Weekly	56
Male	Monthly	54

Findings

- Female donors are more represented in the “Never” and “Weekly” categories.
- Male donors are most represented in the “Yearly” donation category.
- Female donors outnumber male donors in most donation frequency categories.
- Both genders show strong participation across recurring donation frequencies.

Business Insight

Donation frequency patterns differ slightly by gender. Female donors demonstrate stronger representation in regular weekly giving, while male donors are more concentrated among yearly donors. these differences suggest that fundraising campaigns could be tailored according to donor behavior and preferred giving frequency.

Business Question 6: Which Job Fields Have the Highest Average Donation Amounts?

To identify the most valuable donor segments, average donation amounts were analyzed across job fields.

Job Field	Donor Count	Average Donation
Research and Development	84	272.17
Legal	66	262.26
Training	84	258.58
Accounting	80	256.30
Product Management	90	253.31
Services	80	248.23
Human Resources	93	247.96
Marketing	74	246.69
Support	79	246.52
Business Development	94	236.87
Engineering	93	236.22
Sales	83	229.02

Findings

- Research and Development professionals recorded the highest average donation amount (272.17).
- Legal professionals ranked second with an average donation of 262.26.
- Training and Accounting professionals also demonstrated strong donation performance.
- Sales donors recorded the lowest average donation among the analyzed job fields.

Business Insight

Research and Development, Legal, and Training professionals represent high-value donor segments. Targeted fundraising campaigns focused on these groups may generate stronger returns compared to broader, non-segmented campaigns.

Business Question 7: What are the Most Common Car Brands Among Donors?

Car Brand	Donor Count
Ford	91
Chevrolet	78
Toyota	51
Mitsubishi	44
Pontiac	40
GMC	39
Mazda	37
Dodge	36
Mercedes-Benz	36
Volkswagen	36

Findings

- Ford is the most common vehicle brand among donors, with 91 donors.
- Chevrolet ranks second with 78 donors.
- Toyota ranks third with 51 donors.
- The remaining brands have relatively similar donor counts.

Business Insight

The donor population appears to be concentrated around mainstream vehicle brands such as Ford, Chevrolet, and Toyota. This information can support demographic profiling and targeted marketing initiatives.

Business Question 8: What Are the Most Popular Movie Genres Among Donors?

Movie Genre	Donor Count
Drama	158
Comedy	101
Documentary	70
Comedy Drama	46
Drama Romance	42
Comedy Drama Romance	26
Horror Thriller	18
Horror	17
Drama War	15

Findings

- Drama is the most popular movie genre among donors, with 158 donors.
- Comedy ranks second with 101 donors.
- Documentary ranks third with 70 donors.
- Most donor preferences are concentrated around Drama and Comedy-related genres.

Business Insight

Donors appear to prefer storytelling and entertainment-focused genres. These preferences could be useful when designing donor engagement campaigns, promotional content, and personalized communication strategies.

Key Insights

1. Strong Overall Donation Performance

The dataset contains **1,000 donors** who contributed a total of **249,085**, with an average donation of **249.09** per donor. This indicates a healthy level of donor participation and contribution.

2. Male Donors Contributed More Overall

Although the donor population was nearly balance (**508 female** and **492 males**), male donors contributed a higher total donation amount (**127,628**) compared to female donors (**121,457**). Male donors also recorded a higher average donation (**259.41**) than female donors (**239.09**).

3. California Generated the Highest Donations

California ranked first among all states with **113 donors** and a total donation amount of **30,264**. Texas and Florida followed with **24,097** and **20,562** respectively, making these states key fundraising regions.

4. Research and Development Professionals Were the Highest-Values Donors

Among all job fields, **Research and Development** donors recorded the highest average donation among (**272.17**). legal and Training professionals also demonstrated strong donation behavior.

5. Yearly Donations Were the Most Common Donation Pattern

The largest donation frequency category was **Yearly** with **135 donors**, followed by **Never (133)** and **Weekly (129)**. This suggests opportunities to convert infrequent donors into more regular contributors.

6. Donor Preferences Reveal Useful Demographic Patterns

Ford was the most common vehicle brand among donors (**91 donors**), while **Drama** was the most preferred movie genre (**158 donors**). These characteristics can help support donor profiling and audience segmentation strategies.

7. Data Quality Was Generally Good

No duplicate donor records were identified, and most fields contained complete information. The only issue was **459 missing university records**, which may limit certain demographic analyses.

Recommendations

Based on the analysis, the following recommendations are proposed:

1. Focus fundraising efforts on high-performing states such as California, Texas, and Florida, which generated the highest total donation amounts.
2. Develop targeted campaigns for donors working in Research and Development, Human Resources, Legal, and Product Management, as these groups demonstrated strong donation performance.
3. Implement donor retention programs aimed at yearly, weekly, and monthly donors to encourage continued engagement and recurring contributions.
4. Launch re-engagement initiatives targeting inactive donors classified under the “Never” donation frequency category.
5. Consider gender-based donor segmentation strategies, as male donors demonstrated higher average and total donation amounts despite being fewer in number.
6. Improve data collection processes to reduce missing values in the university field and enhance future donor profiling efforts.

Limitations & Assumptions

Limitations

- The university field contained **459 missing values**, limiting educational background analysis.
- The analysis is based solely on the provided dataset and may not represent the entire donor population
- Donor preference such as car brands and movie genres may not directly influence donation behavior.

Assumptions

- All donation records were assumed to be accurate and valid.
- Donor IDs were assumed to uniquely identify each donor since no duplicate IDs were found.
- Missing university values were treated as unavailable information and were not replaced.
- Donation amounts were assumed to be recorded in a consistent format across the dataset.

CONCLUSION

This project successfully analyzed donor and donation data using SQL and PostgreSQL. Through data exploration, quality assessment, cleaning, and business analysis, valuable insights were uncovered regarding donor behavior, donation patterns, geographic trends, and donor demographics.

The analysis revealed that donation performance varies by gender, profession, location, and donation frequency. High-value donor segments were identified, and several opportunities for improving fundraising effectiveness were highlighted.

Overall, the dataset provided meaningful insights that can support data-driven fundraising decisions and help organizations optimize donor engagement and revenue generation strategies.